

User Guide: CDP AI/ML Platform — Production ML Governance

This guide explains how business users and technical users should use the application.

1. What The App Does

The application helps teams review AI/ML decision models, understand their business impact, and inspect the technical evidence behind each run.

It is organized around four model workflows:

- TensorFlow Next Best Action Model
- TensorFlow Churn Propensity Model
- Bayesian Media Mix Optimization
- Causal Incrementality Model

Each workflow moves from model inputs to recommendation, impact forecast, decision evidence, activation output, and governance evidence.

2. Recommended Presentation Flow

Use this sequence for an end-to-end walkthrough:

- Start at AI Decision Portfolio.
- Open AI Impact Summary to show portfolio-level health.
- Open the Model Decision Workbench.
- Walk through model inputs, recommendation, impact forecast, and decision evidence.
- Open Simulation Flow to change setup and run the AI/ML pipeline one stage at a time.
- Open Architecture Review to explain the technical runtime and evidence path.

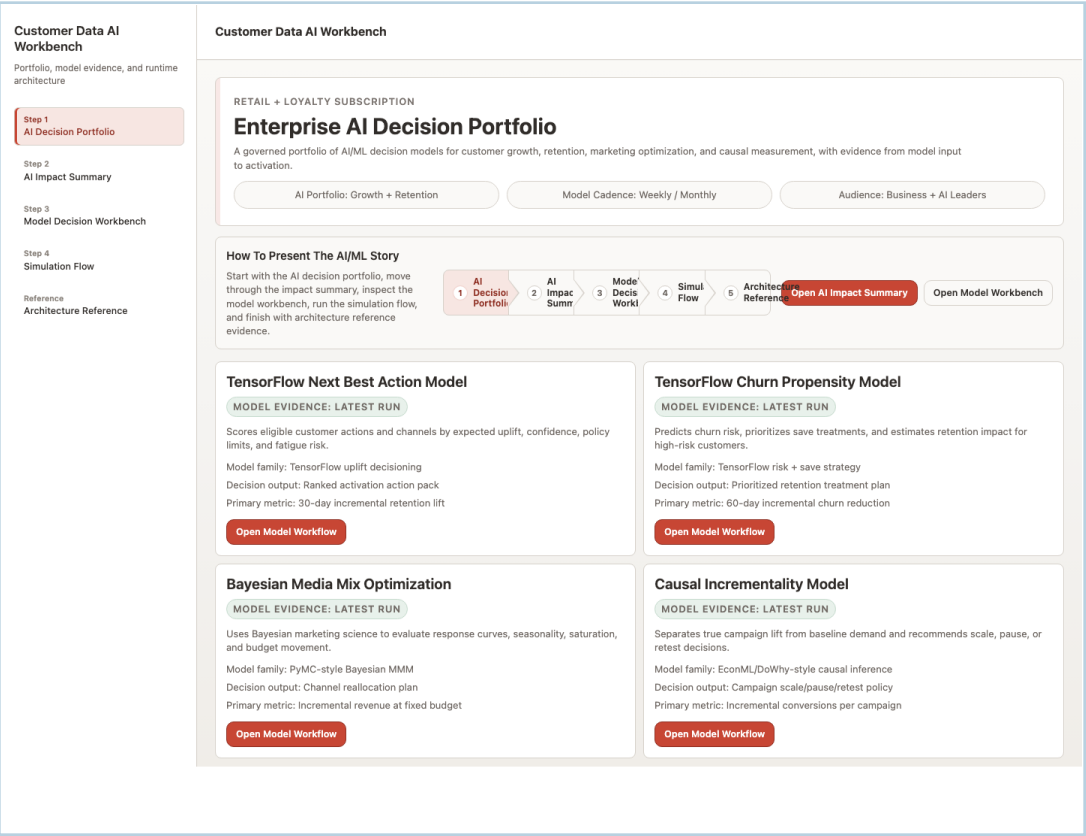
3. Visual Walkthrough

The screenshots below show the recommended flow a user should follow during a business or technical walkthrough.

Flow Step 1: Start With The AI Decision Portfolio

Use this page to introduce the model portfolio and explain the four AI/ML decision workflows.

AI Decision Portfolio



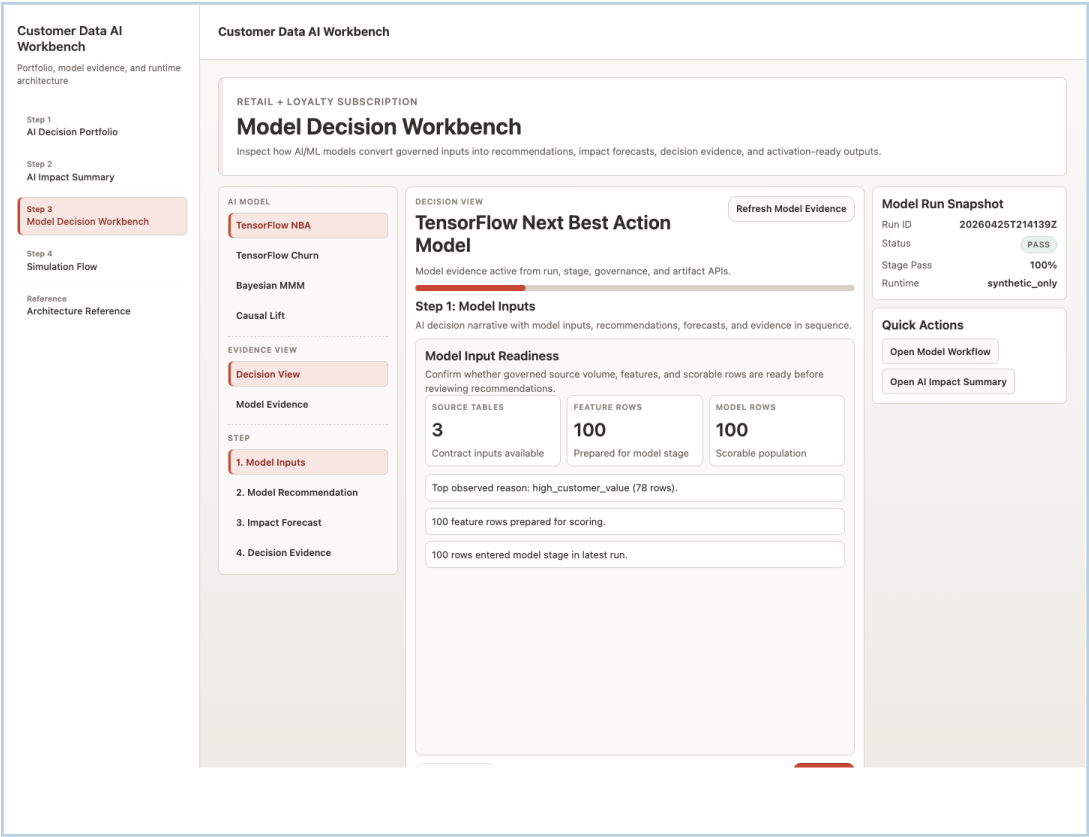
What to point out:

- Four governed AI/ML decision models.
- Each model has a clear business purpose and primary metric.
- The recommended presentation sequence starts with customer-level AI decisioning.

Flow Step 2: Open The Model Decision Workbench

Open a model workflow and start with Model Inputs. This is where users see the data readiness and scorable population before looking at recommendations.

Model Inputs



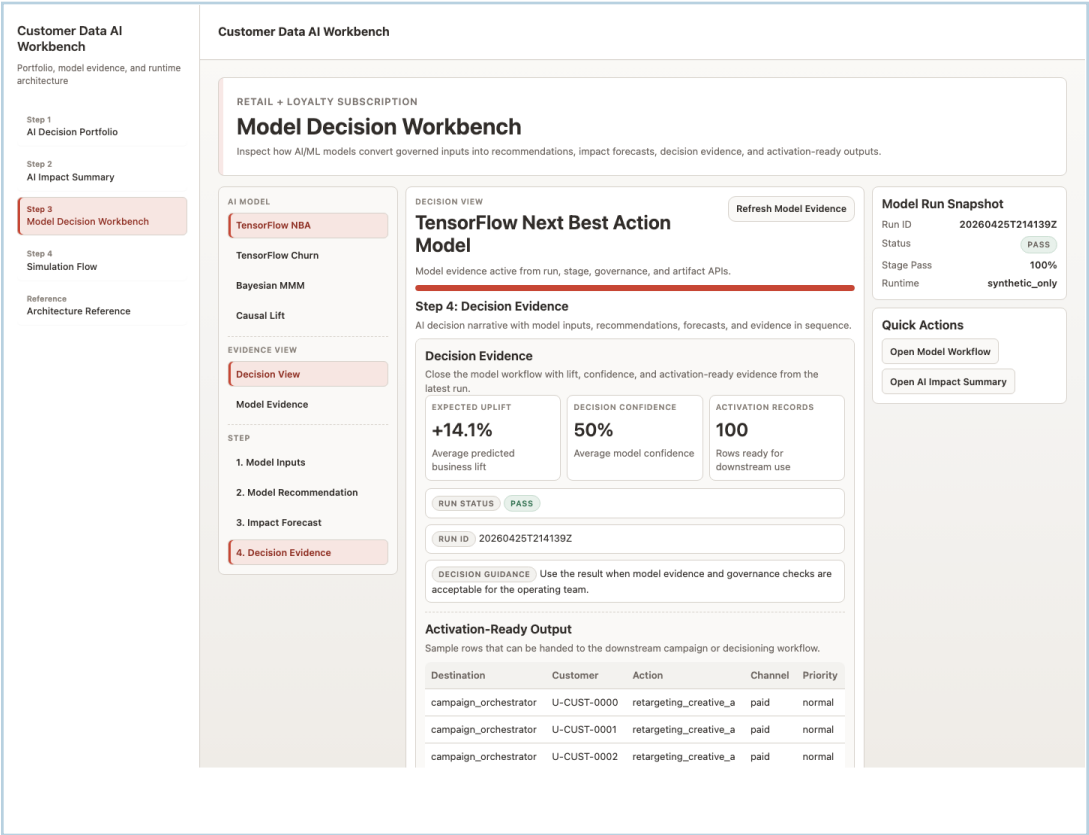
What to point out:

- The left navigation keeps model, evidence view, and step selection visible.
- The center panel explains one decision step at a time.
- The right-side snapshot confirms run ID, status, stage pass rate, and runtime.

Flow Step 3: Review Decision Evidence

Move to Decision Evidence to close the business workflow with uplift, confidence, activation records, and run status.

Decision Evidence



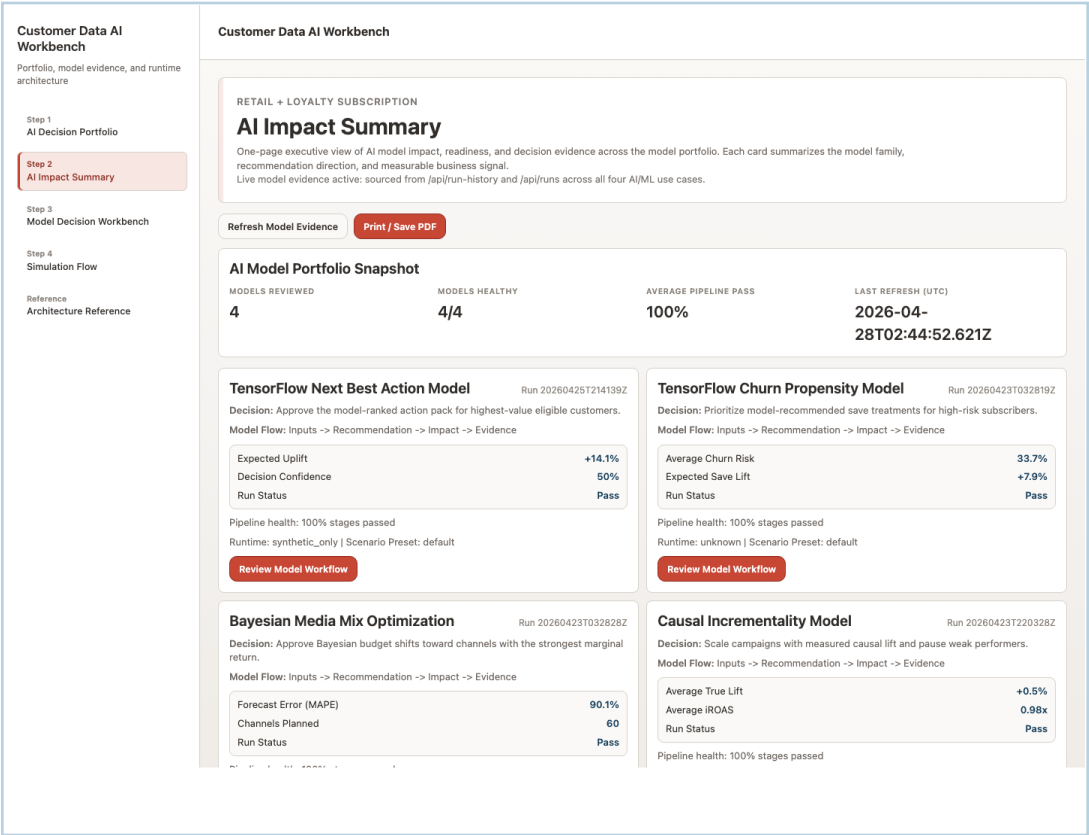
What to point out:

- Expected uplift summarizes predicted business value.
- Decision confidence shows the average model confidence signal.
- Activation records show that outputs are ready for downstream decisioning.

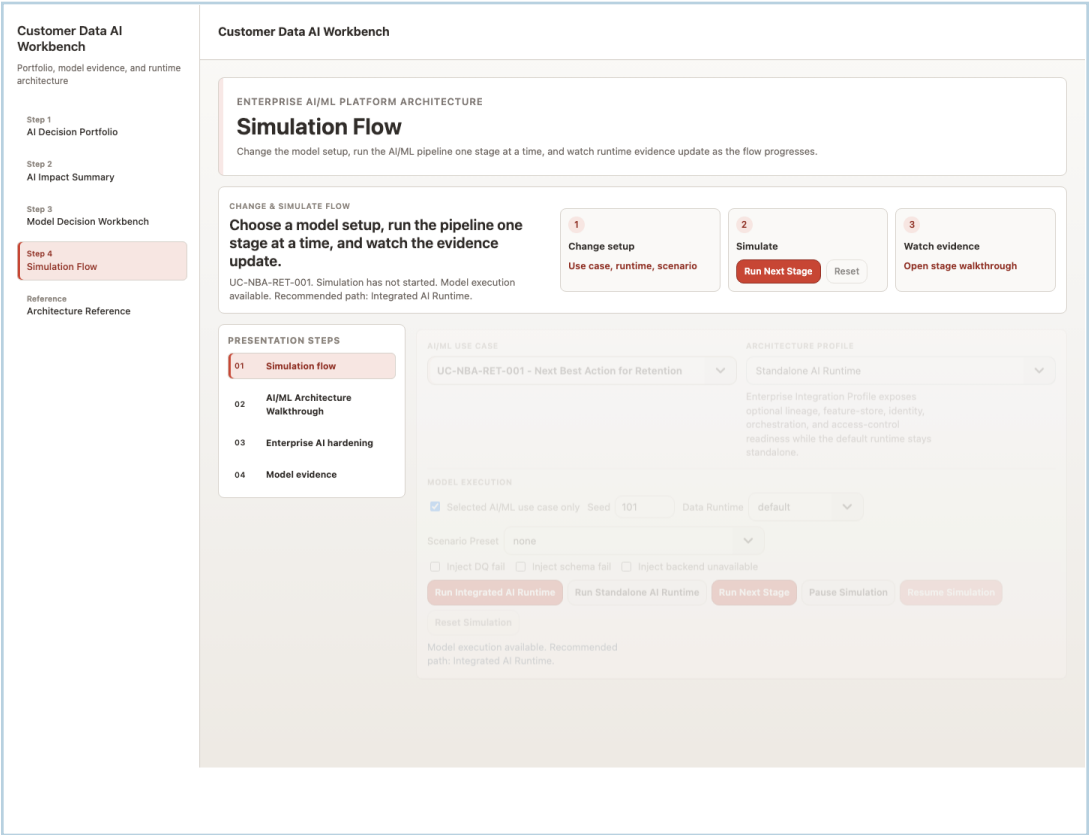
Flow Step 4: Review AI Impact Summary

Open AI Impact Summary when you need a portfolio-level executive view.

AI Impact Summary



Simulation Flow



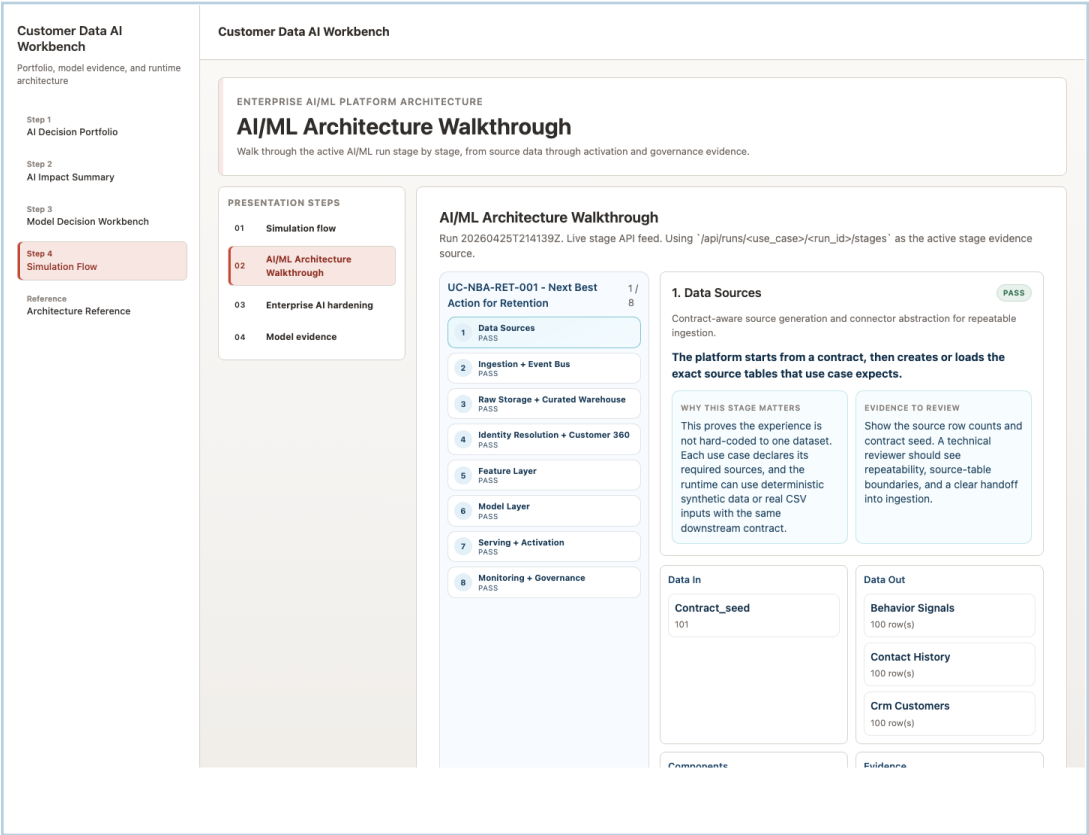
What to point out:

- AI/ML Use Case is a dropdown so users can switch models.
- Model Execution controls expose runtime, scenario, seed, and failure simulation.
- Run Next Stage advances the walkthrough and updates runtime evidence.

Flow Step 6: Walk The AI/ML Architecture

Use AI/ML Architecture Walkthrough to explain the pipeline from Data Sources to Monitoring + Governance.

AI/ML Architecture Walkthrough



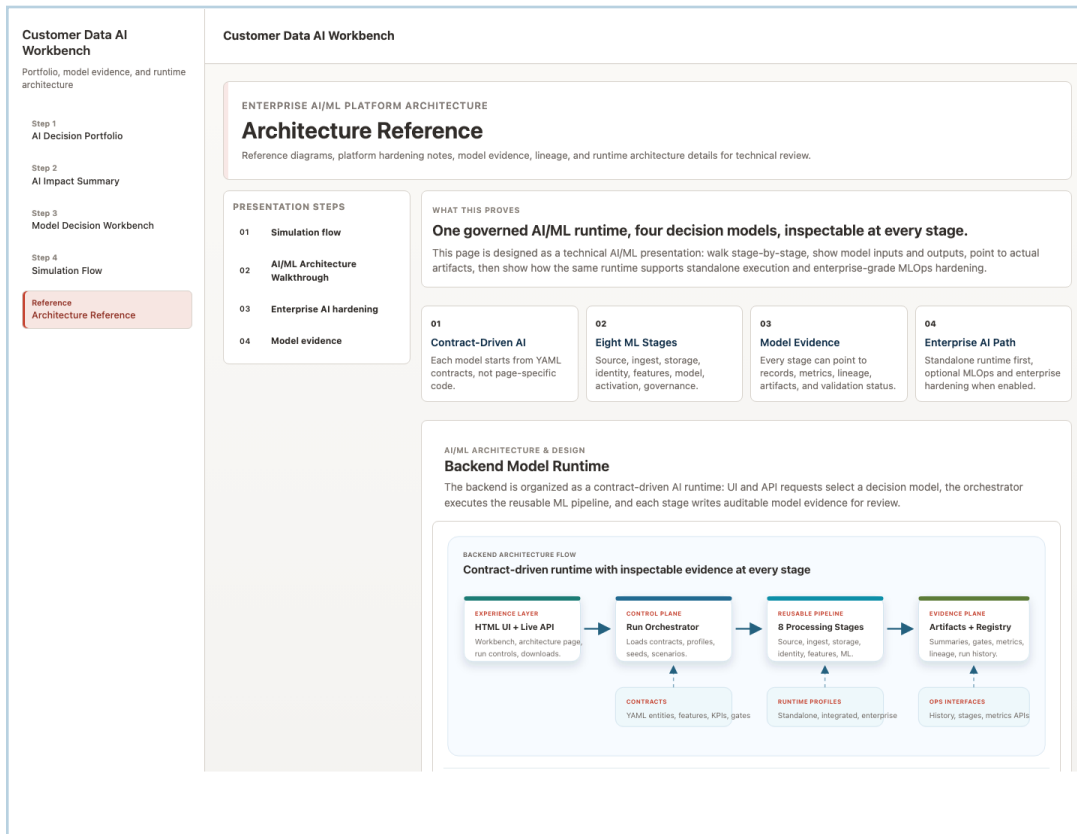
What to point out:

- Each stage has a runtime responsibility.
- Each stage produces artifacts or metrics.
- Technical users can inspect inputs, outputs, components, and evidence.

Flow Step 7: Review Architecture And Evidence

Use Architecture Review and Model Evidence to review platform diagrams, architecture flow, run history, and portfolio health.

Architecture Review And Evidence



What to point out:

- Architecture Review separates platform diagrams from action-oriented simulation.
- Model Evidence shows run artifacts and execution evidence.
- Model Run History compares recent runs.
- AI Portfolio Health summarizes latest run status across use cases.

Flow Step 8: Switch To Technical Model Evidence

In the Model Decision Workbench, switch from Model Decision to Model Evidence for a deeper technical review.

Technical Model Evidence

The screenshot displays the 'Customer Data AI Workbench' interface. On the left, a sidebar lists steps: Step 1 (AI Decision Portfolio), Step 2 (AI Impact Summary), Step 3 (Model Decision Workbench - highlighted), Step 4 (Simulation Flow), and a Reference Architecture Reference. The main content area is titled 'Model Decision Workbench' under the 'RETAIL + LOYALTY SUBSCRIPTION' header. It shows the 'TensorFlow Next Best Action Model' with a 'Refresh Model Evidence' button. Below this, 'Step 3: Responsible AI Evidence' is highlighted, with a sub-header 'Governance + Evidence'. This section includes three cards: 'DATA QUALITY STATUS' (not_applicable), 'BLOCKERS / WARNINGS' (0/0), and 'REGISTRY STAGE' (candidate). A 'Governance Meta' table shows Run Status (pass), Infra (local), Run ID (20260425T214139Z), and Monitoring Report (download). 'Issue Names' shows Blockers: none and Warnings: none. A 'Data Quality Reason' note states: 'Data-quality gates apply to real dataset mode only.' At the bottom, 'Validation Gates' shows a table with two rows: 'uplift_validation_pass' and 'calibration_pass', both with a 'PASS' status.

Customer Data AI Workbench

Portfolio, model evidence, and runtime architecture

Step 1
AI Decision Portfolio

Step 2
AI Impact Summary

**Step 3
Model Decision Workbench**

Step 4
Simulation Flow

Reference
Architecture Reference

Model Decision Workbench

RETAIL + LOYALTY SUBSCRIPTION

Inspect how AI/ML models convert governed inputs into recommendations, impact forecasts, decision evidence, and activation-ready outputs.

AI MODEL

TensorFlow NBA

TensorFlow Churn

Bayesian MMM

Causal Lift

EVIDENCE VIEW

Decision View

Model Evidence

STEP

1. Data + Model Contract

2. Model Runtime + Activation

3. Responsible AI Evidence

MODEL EVIDENCE VIEW

Refresh Model Evidence

TensorFlow Next Best Action Model

Model evidence active from run, stage, governance, and artifact APIs.

Step 3: Responsible AI Evidence

Deep technical walkthrough: contracts, stage evidence, model lineage, and governance artifacts.

Governance + Evidence

Review readiness gates, data-quality outcomes, and artifact auditability before sign-off.

DATA QUALITY STATUS

not_applicable

Mode: synthetic_only

BLOCKERS / WARNINGS

0/0

Must-fix vs review-only issues

REGISTRY STAGE

candidate

Model lifecycle state

Governance Meta

Run Status	pass
Infra	local
Run ID	20260425T214139Z
Monitoring Report	download

Issue Names

Blockers: none

Warnings: none

Data Quality Reason: Data-quality gates apply to real dataset mode only.

Validation Gates

Governance checks emitted by monitoring layer.

Gate	Status
uplift_validation_pass	PASS
calibration_pass	PASS

Model Run Snapshot

Run ID	20260425T214139Z
Status	PASS
Stage Pass	100%
Runtime	synthetic_only

Model Evidence Snapshot

Components	8/8 pass
Artifacts	17/28 present
Data Quality	not_applicable (0/0)
Model Backend	tensorflow

Quick Actions

Open Model Workflow

Open AI Impact Summary

What to point out:

- Contract controls show use case, runtime, scenario, and model constraints.
- Technical evidence connects components, APIs, stage outputs, and artifacts.
- The same workflow can be explained to business users or technical users without changing apps.

4. Starting The App

From the repository root:

```
make standalone
```

Open:

```
http://127.0.0.1:8080/ui/experience/index.html
```

If the environment is already prepared:

```
make standalone-fast
```

For live UI/API mode:

```
make ui-live
```

5. Main Pages

Page | URL | Primary Audience

AI Decision Portfolio | /ui/experience/business-home.html | Business and executive users

AI Impact Summary | /ui/experience/business-exec-summary.html | Executives and business owners

Model Decision Workbench | /ui/experience/workbench.html | Business users and technical reviewers

Simulation Flow | /ui/experience/index.html#simulate | Architects, ML engineers, and platform teams

Architecture Review | /ui/experience/index.html#overview | Architects, ML engineers, and platform teams

6. Business User Guide

Business users should focus on the decision story and model impact.

Step 1: Open AI Decision Portfolio

Use the portfolio page to understand which models are available and what each model is designed to decide.

Look for:

- Model name
- Decision output

- Primary business metric
- Recommended presentation sequence

Step 2: Open A Model Workflow

Click Open Model Workflow for one of the models.

Recommended first workflow:

TensorFlow Next Best Action Model

Step 3: Walk The Business Steps

In the Model Decision Workbench, use the left-side step navigation:

- Model Inputs
- Model Recommendation
- Impact Forecast
- Decision Evidence

Use these questions while presenting:

- What customer, campaign, or spend signals entered the model?
- What does the model recommend?
- What impact does the model forecast?
- What evidence supports the decision?

Step 4: Review The AI Impact Summary

Open AI Impact Summary to see portfolio-level model health.

Look for:

- Models reviewed
- Models healthy

- Average pipeline pass rate
- Latest run status
- Model-specific business KPIs

Step 5: Share Or Print

Use Print / Save PDF on the AI Impact Summary page when you need an executive-ready output.

7. Technical User Guide

Technical users should focus on architecture, run evidence, artifacts, and governance.

Step 1: Open Simulation Flow

Use Simulation Flow to change the model setup and run the end-to-end platform one stage at a time.

The core architecture is:

Data Sources

- > Ingestion + Event Bus
- > Raw Storage + Curated Warehouse
- > Identity Resolution + Customer 360
- > Feature Layer
- > Model Layer
- > Serving + Activation
- > Monitoring + Governance

Step 2: Walk The AI/ML Architecture

Use AI/ML Architecture Walkthrough to walk stage by stage.

For each stage, review:

- What the stage does
- Data in
- Data out
- Components
- Evidence artifacts

Step 3: Use Model Evidence View

Open Model Decision Workbench and select Model Evidence.

Use this view to inspect:

- Contract controls
- Stage inputs and outputs
- Model backend
- Model metrics
- Activation payloads
- Validation gates
- Artifact paths

Step 4: Inspect Architecture Review And Model Evidence

Open Architecture Review for platform diagrams, then open Model Evidence for run artifacts and history.

Review:

- Architecture Review
- Model Evidence
- Model Run History
- AI Portfolio Health

Step 5: Run A New Model Workflow

Use Model Execution controls on Simulation Flow.

Common options:

- Select an AI/ML use case.
- Choose Standalone AI Runtime or Integrated AI Runtime.
- Choose runtime mode.
- Select a scenario preset if needed.
- Click Run Standalone AI Runtime or Run Integrated AI Runtime.

Step 6: Simulate Stage Execution

Use the simulation controls to explain the pipeline gradually:

- Run Next Stage
- Pause Simulation
- Resume Simulation
- Reset Simulation

This is useful when presenting the architecture to technical audiences.

8. Runtime Profiles

Profile | Use When

Standalone AI Runtime | You want the app to run locally without external services.

Integrated AI Runtime | You want to show optional Kafka, Postgres, or MinIO-style infrastructure mirroring.

Enterprise Integration Profile | You want to show readiness for MLflow, Feast, Splink, Airflow, Keycloak, monitoring, and metrics.

The app can run standalone without deployment dependencies. Enterprise integrations are optional.

9. Common Tasks

Run all model workflows

```
python3 -m pipelines.cli run-stack-all
```

Run one workflow

```
python3 -m pipelines.cli run-stack --use-case UC-NBA-RET-001
```

Run synthetic-only mode

```
python3 -m pipelines.cli run-stack --use-case UC-NBA-RET-001 --runtime-mode  
synthetic_only
```

View recent runs

```
python3 -m pipelines.cli list-runs
```

Inspect run stages

```
python3 -m pipelines.cli show-stages --use-case UC-NBA-RET-001 --run-id <run_id>
```

Inspect model readiness

```
python3 -m pipelines.cli model-readiness --use-case UC-NBA-RET-001 --run-id <run_id>
```

10. Troubleshooting

If the app is not loading:

- Confirm the server is running.
- Confirm the URL uses the right port.
- Run make standalone-fast if dependencies and artifacts already exist.
- Run make smoke-standalone to validate the app.

If model evidence is missing:

- Run at least one workflow.
- Refresh the page.
- Check artifacts/<use_case_id>/<run_id>/summary.json.

If integrated runtime is unavailable:

- Use Standalone AI Runtime.
- Confirm optional infrastructure is running before using Integrated AI Runtime.

11. What To Show In A Meeting

For business stakeholders:

- AI Decision Portfolio
- One Model Workflow
- AI Impact Summary

For technical stakeholders:

- AI/ML Platform Architecture
- Eight ML stages
- Model Evidence View
- Model Run History
- Runtime Evidence

For platform leaders:

- Business value from model workflows
- Evidence-backed architecture
- Standalone runtime
- Enterprise integration path